Proof and Induction WS 2

1. Use a direct proof to prove each of the following statements.

    a. The product of two odd integers I odd.

    b. The sum of two consecutive odd numbers is divisible by 4.

2. Use a contrapositive proof to prove each of the following statement: If a and b are integers and ab is even, then at least one of a and b is even.

3. Use a proof by contradiction to prove each of the following statement: $\sqrt{5}$ is irrational.

4. Prove each of the following logical equivalence: Let n be a positive integer. $n-3$ is odd if and only if $n+2$ is even.

5. Prove that the number $1-5\sqrt{2}$ is irrational.

6. It is known that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square. Use this result to help prove that $\sqrt{6}+\sqrt{10}$ is irrational.

7. Write each sentence in mathematical notation.

    a. For all positive real numbers x and y, if x is greater than y, then the square of x is greater than the square of y.

    b. There exists a rational number c between integers a and b where $a<b$, such that c is the average of a and b.

    c. Let n be a positive integer such that for all n, $1+2+3+\cdots+n=\frac{n(n+1)}{2}$.

8. Prove by contradiction that $\log_5 13$ is an irrational number.

9. It is given that $a>0, b>0$ are real numbers. Consider the statement $\forall a\left(\forall b, \log_{\frac{1}{a}}\frac{1}{b}=\log_a b\right)$. Prove that the statement is true.

10. Use proof by contradiction to show that $\log_2 5$ is irrational.

11. It is given that p and q are both real numbers. Consider the statement $\forall p\left(\forall q, \frac{1}{p^2}<\frac{1}{q^2}\right)$. Either prove that the statement is true or provide a counterexample.

12. If $u_1=0, u_2=9$ and $u_n=6u_{n-1}-9u_{n-2}$ for $n\geq 3$, show that $u_n=(n-1)3^n$ for $n\geq 1$.

13. If $u_1 = 8, u_2 = 20$ and $u_n = 4u_{n-1} - 4u_{n-2}$ for $n \geq 3$, show that $u_n = (n+3)2^n$ for $n \geq 1$.

14. Prove by Mathematical Induction that for $n \geq 1, \frac{1}{1!} + \frac{2}{3!} + \frac{3}{5!} + \cdots + \frac{n}{(2n-1)!} \leq 2 - \frac{1}{(2n)!}$.

15. If $u_1 = 5$, $u_2 = 13$ and $u_n = 5u_{n-1} - 6u_{n-2}$ for $n \geq 3$, show that $u_n = 2^n + 3^n$ for $n \geq 1$

16.

a. Show that for $n \geq 1, \frac{d^n}{dx^n} \ln(1+x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}$

b. Show that for $n \geq 1, \frac{d^n}{dx^n} \ln(1-x) = \frac{(-1)(n-1)!}{(1-x)^n}$

17. Suppose that $H(n) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$. Prove by induction that $n + H(1) + H(2) + H(3) + \cdots + H(n-1) = n\,H(n)$, for all positive integer values of n.

18. A sequence $\{t_n\}$ is defined so that $t_{n+2} = 6t_{n+1} - 5t_n$ and $t_1 = 2, t_2 = 6$. Prove that $t_n = 5^{n-1} + 1$.

19. Prove by induction that $2n^3 - 3n^2 + n + 31 \geq 0$.

20. If $u_n = 2^{n+2} + 3^{2n+1}$, show that $u_{n+1} = 2u_n + 7 \times 3^{2n+1}$, and hence show that u_n is divisible by 7 for $n \geq 1$.

21. If $u_1 = 2, u_2 = 3$ and $u_{n+2} = 3u_{n+1} - 2u_n$, prove by induction that $u_n = 2^{n-1} + 1$.

22. If $u_n = 5^{2n} + 3n - 1$, show that u_n is divisible by 9 for $n \geq 1$.